SmartGuard ESP32 – Voltage and Overcurrent Protector

Project Idea

As project requirement, a voltage and overcurrent protection system is developed using an ESP32. The main idea was to monitor voltage and current continuously and disconnect the load whenever the values go beyond the selected limits. Here, Wokwi website has been used for the simulation because I wanted to test the protection logic without working directly with AC mains voltage.

How I Designed the System

Two slide potentiometers have been used as inputs. One is used to represent voltage from 0 to 300 V, and the other represents current from 0 to 10 A. These are simulated values, not actual AC measurements. The ESP32 reads the values through GPIO 33 and GPIO 32. I connected a relay module to GPIO 23, a red LED to GPIO 19, and a reset button to GPIO 18. The green LED on the relay output represents whether the load is connected. Initially the protection limits have been kept at 200–250 V and 5 A. When voltage is too low, too high, or current exceeds the limit, the ESP32 turns the relay OFF. I also added fault latching, so the load does not turn ON automatically when the input becomes normal again. The user has to reset it after the condition becomes safe.

Computer Communication

At first, the Serial Monitor has been used to check readings and change the protection limits using commands like SET_OV, SET_UV, and SET_OC.

Later, I added MQTT communication and built a simple web dashboard using HTML, CSS, and JavaScript. I also used a Node.js backend to receive the ESP32 data and send settings commands. Now I can change the voltage and current limits from the dashboard and see the updated values in the Wokwi Serial Monitor. I kept the physical reset button for reconnecting the load.

Results and faced Problems

I tested normal operation, undervoltage, overvoltage, overcurrent, fault latching, and reset. During development, I had some problems with the relay logic and MQTT communication. I tested the relay separately to confirm which signal turns it ON. I also had to fix errors in the backend and dashboard before the settings could update properly. Finally, I checked the dashboard readings against the Wokwi Serial Monitor and tested whether changing the limits from the dashboard was reflected in the ESP32.

Limitations

This is a simulation-based prototype. I used potentiometers instead of actual voltage and current sensors, and the load is represented by an LED. The MQTT setup uses a public broker for demonstration, so the remote settings feature is not secure enough for a real electrical protection device. The system would need proper isolated sensors, protection hardware, secure communication, and more testing before real-world use.